

Replication Study of OSTI 2469515: “Supervised Extraction of Near-Complete Genomes from Multiple Metagenomes”

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2026-04-24

Abstract

We attempt a partial replication of Parrello *et al.* (2021, OSTI 2469515), which describes PATRIC’s reference-guided (“supervised”) metagenome binning pipeline based on **pheS** anchors, discriminating protein 12-mers, and **EvalG/EvalCon** quality scoring. The authors’ pipeline relies on the proprietary SEEDtk framework and the 193,980-genome PATRIC/BV-BRC reference database, neither of which is practical to install or stage on a single 10-core iMac in a 2.5 hour window. We therefore replicate the paper’s **baseline comparator** (MetaBAT2 on a SPAdes co-assembly with per-sample coverage from minimap2/samtools) on a controlled synthetic 5-species metagenome generated with InSilicoSeq. Because ground truth is known, we score completeness and contamination directly by aligning each bin’s contigs to the source reference genomes. All five expected genomes are recovered as 100%-pure bins with 85.6–98.3% completeness, and all five pass the MIMAG high-quality threshold (completeness $\geq 80\%$, contamination $\leq 10\%$). This is consistent with the paper’s observation that MetaBAT2 performs well when differential coverage information across samples is available, and it demonstrates that the non-supervised comparator recovers the “easy” portion of the bin yield with near-perfect precision on clean synthetic data. The paper’s core claim — that reference-guided binning *exceeds* MetaBAT2 on complex real human-gut samples (205 vs. 180 HQ bins across 23 ERR/SRR samples) — cannot be replicated here without SEEDtk/PATRIC and is documented as out-of-scope.

1 Paper Summary (Claims Relevant to Replication)

- **Method:** Reference-guided binning. **pheS** (phenylalanyl-tRNA synthetase α) sequences on assembled contigs are BLASTed against PATRIC. Each pheS-bearing contig ≥ 400 bp with $\geq 4\times$ coverage seeds a bin; contigs are then assigned based on sharing ≥ 10 *discriminating protein 12-mers* with the bin’s reference. Unbinned contigs are rescued via ≥ 50 bp exact DNA matches.
- **Quality scoring:** **EvalG** (completeness/contamination, CheckM-like) and **EvalCon** (fine consistency via random-forest predictors over $\sim 1,300$ reliably predictable gene roles, $> 93\%$ accuracy).
- **HQ threshold:** contamination $\leq 10\%$, fine consistency $\geq 87\%$, completeness $\geq 80\%$, exactly one pheS of valid length.
- **Benchmark:** 23 human metagenomic samples from Pasolli 2019.
 - PATRIC supervised: 205 good / 164 poor bins

- MetaBAT2 + Bowtie2: 180 good / 660 poor bins
- MetaBAT2 (contigs only): 122 good / 431 poor bins
- **Postprocessing finding:** Contamination-driven role filtering raised the average HQ bins/sample from 1.7 to 8.17.
- **pheS–genome similarity Pearson** $r = 0.97$; no HQ bins retrieved when best reference pheS similarity drops below 0.3.

2 What We Replicated vs. Skipped

Component	Disposition
Read simulation	Replicated with InSilicoSeq (HiSeq error model).
Co-assembly	Replicated with metaSPAdes 4.2.0.
Per-sample mapping	Replicated with minimap2 sr preset + sam-tools.
Baseline binning (MetaBAT2)	Replicated.
Ground-truth quality scoring	Done directly via minimap2 asm5 alignment of bin contigs to reference genomes (stronger than CheckM because we have the exact sources).
CheckM/CheckM2	Skipped: osx-64 install friction; not needed given ground truth. Equivalent MIMAG thresholds applied.
PATRIC supervised pipeline (SEEDtk)	Out of scope: requires the 193,980-genome PATRIC database, RASTtk, EvalG, EvalCon, and SEEDtk — not installable on macOS in the time budget without cloud/BV-BRC access.
pheS anchoring / discriminating 12-mers	Not reimplemented; comparator only.
Real human-gut benchmark (23 Pasolli samples)	Substituted with a 5-species synthetic community (~ 1 GB reads) to fit compute budget.

3 Methods

3.1 Synthetic Community

Five diverse bacterial reference genomes were downloaded from NCBI RefSeq (*Escherichia coli* K-12 MG1655, *Staphylococcus aureus* NCTC 8325, *Bacillus subtilis* 168, *Lactobacillus gasseri* ATCC 33323, *Porphyromonas gingivalis* ATCC 33277; total ≈ 15.9 Mb). Contig headers were rewritten with a species prefix for ground truth tracking.

Two metagenomic “samples” of 10^6 read pairs (2×126 bp) each were generated with InSilicoSeq 2.0.1 (HiSeq error model, 8 CPUs) with deliberately skewed and *opposite* abundance vectors:

Species	Sample1	Sample2
<i>E. coli</i>	0.50	0.10
<i>S. aureus</i>	0.05	0.40
<i>B. subtilis</i>	0.25	0.10
<i>L. gasseri</i>	0.10	0.30
<i>P. gingivalis</i>	0.10	0.10

This differential-abundance design is what allows MetaBAT2’s tetranucleotide plus coverage signal to separate species cleanly.

3.2 Pipeline

1. **Co-assembly:** metaSPAdes 4.2.0, `--meta --only-assembler`, on the concatenated PE library (both samples). Output filtered to contigs ≥ 1000 bp.
2. **Mapping:** minimap2 2.30 `-ax sr` per sample, piped to `samtools sort`. $> 98.7\%$ reads-well-mapped per sample.
3. **Coverage:** `jgi_summarize_bam_contig_depths` across both BAMs.
4. **Binning:** MetaBAT2 2.18, `-m 1500 --seed 42`.
5. **Evaluation:** Each bin’s contigs aligned to the combined reference with minimap2 `-x asm5`. Each query contig is assigned to the species with the most matching bases; per-species totals give purity (contamination = $1 - \text{purity}$), and completeness is the ratio of assigned bases to that species’ genome length.

Software note: megahit 1.2.9 from bioconda segfaulted on this Intel iMac (both the `count` and `assemble` sub-steps with SIGSEGV irrespective of BMI2 core selection); we switched to metaSPAdes, which ran to completion.

4 Results

metaSPAdes produced 659 contigs totaling 15.58 Mb; 258 contigs ≥ 1000 bp (15.53 Mb) were fed to MetaBAT2. MetaBAT2 produced **5 bins** containing 98.00% of the large-contig bases (15.21 Mb). Per-bin evaluation against the ground truth:

Bin	Species	Purity %	Complete %	Bin size (bp)	Aligned (bp)	MIMAG HQ?
bin.1	<i>P. gingivalis</i>	100.00	85.64	2,021,268	2,016,788	
bin.2	<i>E. coli</i>	100.00	97.19	4,513,831	4,511,214	
bin.3	<i>S. aureus</i>	100.00	98.30	2,775,039	2,773,263	
bin.4	<i>L. gasseri</i>	100.00	95.51	1,809,639	1,809,248	
bin.5	<i>B. subtilis</i>	100.00	96.93	4,090,423	4,086,144	

Headline numbers.

- 5 / 5 genomes recovered.

- Contamination: 0% in every bin (every mapped base in every bin originated from that bin’s main species).
- Completeness range: 85.6%–98.3%; mean 94.7%.
- HQ bins ($\text{MIMAG} \geq 80\%$ / $\leq 10\%$): **5/5**.
- Paper’s HQ threshold ($\text{EvalG} \geq 80\%$ complete, $\leq 10\%$ contam): all 5 would qualify; the EvalCon fine-consistency criterion cannot be checked without RASTtk.

5 Comparison With the Paper

Method	Samples	HQ bins	HQ/sample	Total bins
PATRIC supervised (paper)	23	205	8.91	370
MetaBAT2 + Bowtie2 (paper)	23	180	7.83	840
MetaBAT2, contigs only (paper)	23	122	5.30	553
This work (MetaBAT2, synthetic)	1	5	5.00	5

Our per-sample HQ yield (5) is *at the low end* of the paper’s MetaBAT2 numbers only because our synthetic community contains only 5 species — the pipeline recovered *all* of them. The paper’s samples are real Pasolli human gut metagenomes, in which hundreds of species compete, only a fraction are assembleable from $\sim 10^7$ reads, and MetaBAT2 produces 840 total bins of which only 180 are HQ (21.4% good-bin rate). Our good-bin rate is 100% because there are no low-abundance under-assembled species to over-fragment.

6 What Was and Was Not Reproduced

Reproduced qualitatively.

- MetaBAT2 is capable of producing high-quality bins when differential coverage is informative (confirmed: 5/5 pure HQ bins from 2 samples with opposite abundance vectors).
- The general recipe (read simulate $\rightarrow$ co-assemble $\rightarrow$ map per sample $\rightarrow$ jgi depth $\rightarrow$ MetaBAT2) works end-to-end on a commodity Mac.

Not reproduced (honest accounting).

- The paper’s central contribution — *supervised* binning via pheS + discriminating 12-mers against PATRIC — was *not implemented*. The SEEDtk stack requires a custom RASTtk / SEED / EvalCon install and the full PATRIC genome corpus; this is BV-BRC-hosted and not trivially reproducible off-platform.
- The paper’s head-to-head superiority of PATRIC over MetaBAT2 on real human-gut data was not directly tested; we only reproduced the comparator arm.
- Postprocessing’s claimed boost from 1.7 to 8.17 HQ bins/sample (a $4.8\times$ improvement attributable specifically to EvalCon-driven role filtering) is inaccessible without EvalCon.
- The pheS-similarity / HQ-yield monotone relationship (Table 2 of the paper) is not testable without a reference pheS database and BLAST against PATRIC.

7 Self-Rated Replication Score

4/10. We reproduced the comparator baseline end-to-end and verified it works correctly on controlled data, but we did not reproduce the paper’s actual supervised method or the human-gut benchmark it was measured on. A fairer 8+/10 would require either BV-BRC platform access or a substantial ($\gg 2.5$ h) re-implementation of SEEDtk against a staged PATRIC snapshot.

8 Reproducibility Bundle

All files are under `~/Dropbox/REPLICATE-PROJECT/2469515-Supervised-extraction-of-near-complete-genom`

- `data/` — reference genomes, abundance files, simulated reads
- `assembly/spades/` — metaSPAdes output; `contigs.fa` (filtered)
- `bins/sample{1,2}.bam`, `depth.txt` — alignments and coverage
- `bins/metabat/bin.{1..5}.fa` — final MAGs
- `checkm/bin_eval.tsv` — ground-truth completeness/purity table
- `report/report.pdf` — this document

Environment. conda env `metabin`: metaSPAdes 4.2.0, minimap2 2.30, samtools 1.23.1, metabat2 2.18, seqkit, InSilicoSeq 2.0.1, biopython 1.87, Python 3.13. Host: CherryRd, macOS, 20 CPU / 128 GB RAM.